

UCT702: IT Workshop Skylab/ MATLAB

L	T	P	Cr
1	0	2	2.0

Course Objectives: To provide an understanding of programming concepts and skills needed for basic problem solving using MATLAB, along with specific focus on image processing algorithms.

Introduction to MATLAB

History, basic features, strengths and weaknesses, good programming practices and plan your code.

Working with variables, workspace and miscellaneous commands

Creating variables, overwriting variable, error messages, making corrections, controlling the hierarchy of operations or precedence, controlling the appearance of floating point number, managing the workspace, keeping track of your work session, entering multiple statements per line, miscellaneous commands.

Matrix, array and basic mathematical functions

Matrix generation, entering a vector, entering a matrix, matrix indexing, colon operator, linear spacing, creating a sub-matrix, dimension, matrix operations and functions matrix generators, special matrices, array and array operations, solving linear equations, other mathematical functions.

Basic plotting

Overview, creating simple plots, adding titles, axis labels, and annotations, multiple data sets in one plot, specifying line styles and colours.

Introduction to programming

Introduction, M-File Scripts, script side-effects, M-File functions, anatomy of a M-File function, input and output arguments, input to a script file, output commands.

Control flow and operators

``if ... end" structure, relational and logical operators, ``for ... end" loop, ``while ... end" loop, other flow structures, operator precedence, saving output to a file.

Debugging M-files

Debugging process, preparing for debugging, setting breakpoints, running with breakpoints, examining values, correcting and ending debugging, correcting an M-file.

Laboratory

Implementation of various Image Processing Algorithms.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

Upon completion of this course, students will be able to:

1. Understand the fundamental programming concepts in MATLAB such as variables, data types, control structures, operators, workspace etc.
2. Comprehend the use of matrix, array and basic mathematical functions for problem solving.
3. Understand and implement the M-file scripts and functions.
4. Illustrate and implement various image processing techniques.

Text Books:

1. Gonzalez R. C., Woods R. E., Eddins S. L. , Digital Image Processing using MATLAB, Pearson Education India (2004), 2nd ed.
2. Attaway S., MATLAB: A Practical Introduction to Programming and Problem Solving, Butterworth-Heinemann (2017), 4th ed.

Reference Books:

1. <https://www.mathworks.com/content/dam/mathworks/mathworks-dot-com/moler/exm/book.pdf>
2. https://www.mathworks.com/help/releases/R2014b/pdf_doc/matlab/getstart.pdf